Introduction to Public Economics

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Government Intervention in the Economy

Public Economics Definition

Public Economics (or public sector economics or public finance) = Study of the role of government in the economy

Government is instrumental in most aspects of economic life:

1. Government in charge of huge **regulatory** structure
2. **Taxes:** governments in advanced economies collect 30-50% of National Income in taxes
3. **Expenditures:** tax revenue funds **traditional public goods** (infrastructure, public order and safety, defense) and **welfare state** (Education, Retirement benefits, Health care, Income Support)
4. Macro-economic **stabilization** through central bank (interest rate, inflation control), fiscal stimulus, bailout policies

$\Rightarrow$ We pool a large share of our incomes through government

Why Public Sector Economics is so Important

Public sector economics provides two things:

(a) Empirical evidence on how policies affect who gets what

(b) Theoretical tools for converting empirical evidence and social values into policy implications

Policy choices matter.

- Here's President Biden: **[web link]**.
- Here's President Reagan (first minute): **[web link]**.
- Here's Professor Darrick Hamilton: **[web link]**.
- Here's Ambassador Nikki Haley (first minute): **[web link]**.

Poll 1: The Role of Government

Governments can provide physical order, enhance individual liberty, and promote economic equality to varying degrees. How would you rank those goals in order of importance?

A. Order. Liberty. Equality.
B. Order. Equality. Liberty.
C. Liberty. Order. Equality.
D. Liberty. Equality. Order.
E. Equality. Liberty. Order.

Poll 2: Government and Income Inequality

- Some people think that the government in Washington ought to reduce the income differences between the rich and the poor, perhaps by raising the taxes of wealthy families or by giving income assistance to the poor. Others think that the government should not concern itself with reducing this income difference between the rich and the poor.
- Here is a card with a scale from 1 to 7. Think of a score of 1 as meaning that the government ought to reduce the income differences between rich and poor, and a score of 7 meaning that the government should not concern itself with reducing income differences. What score between 1 and 7 comes closest to the way you feel?

Poll 3: How to Help Lower Earners

Which type of assistance to lower earners would you prefer?

A. A "universal basic income" that gives large cash assistance to those with zero earnings and progressively less higher up the income ladder.

B. An "earning income tax credit" that gives large cash assistance to those with $15,000 of earnings and progressively less at lower and higher earnings.

C. No cash assistance. Instead, the government directly provides all services to lower earners: housing, food, health care, etc.

Poll 4: Health Care Provision

Should the government fund health care for all Americans?

A. Yes

B. No

Poll 5: Medicare for All Funding

Can the government pay for universal healthcare by taxing the top 1%?

A. Yes

B. No

Poll 6: School Choice

Parents pay for public school tuition through their property taxes.
Should parents be able to send those tax dollars to whatever school they send their child to, such as for tuition at a private religious school?

A. Yes

B. No

Poll 7: Climate Change

Which statement best sums up your view of climate change?

A. Society should sacrifice immense amounts in order to counter climate change.
B. Society should sacrifice little in order to counter climate change.
C. I truly don't know what society should do.

Poll 8: Paying for Government

Which statement best sums up your view on taxes and spending?

A. America should not worry about paying for public expenditures and should instead borrow the money.

B. America should borrow money during emergencies like recessions and wars but should otherwise pay for public expenditures with taxes.

C. America should pay for its public expenditures with taxes every year and never borrow.

Policymakers Seek Economists' Help

Bigger View on Government

- Economists have a narrow minded view of individual behavior: purely selfish and economically rational interacting through markets $\Rightarrow$ Limitation to fully understand **public economics**
- Social interactions are critical for humans: we naturally cooperate at many levels: families, communities, nation states, global treaties with very strong/versatile in-group attachments
- Archaic human societies depended on social cooperation for protection and taking care of the young, sick, and old
- $\Rightarrow$ May explain why our modern nation states provide defense and education, health care, and retirement benefits
  Humans may reveal their social nature from the size and nature of their "governments" (informal and formal)

More Modest Role for Economists

Replacing social institutions by markets does not always work:

- Education is primarily government funded: student loans work in economic theory but in practice end up being a huge lifetime burden. For-profit education has a tendency to become a scam
- Retirement benefits: Saving for your own retirement works in theory but in practice most people unable to do so unless institutions (employers/government) help them
- Health care: Health care relies heavily on government/community support everywhere. People are not able to afford or shop rationally for health care

Economists can still play a useful role in understanding when markets can help and how individualistic forces can undermine institutions

Four Questions of Public Finance

1. When should the government intervene in the economy?
2. How might the government intervene?
3. What is the effect of those interventions on economic outcomes?
4. Why do governments choose to intervene in the way that they do?

When Should the Government Intervene in the Economy?

- **1) Market Failures**: Market economy sometimes fails to deliver an outcome that is efficient $\Rightarrow$ Government intervention may improve the situation
- **2) Redistribution**: Market economy generates substantial inequality in economic resources across individuals $\Rightarrow$ Government intervention may help reduce inequality by redistributing resources through taxes and transfers

- First part of the class focuses on Redistribution
- Second part of the class focuses on Market Failures

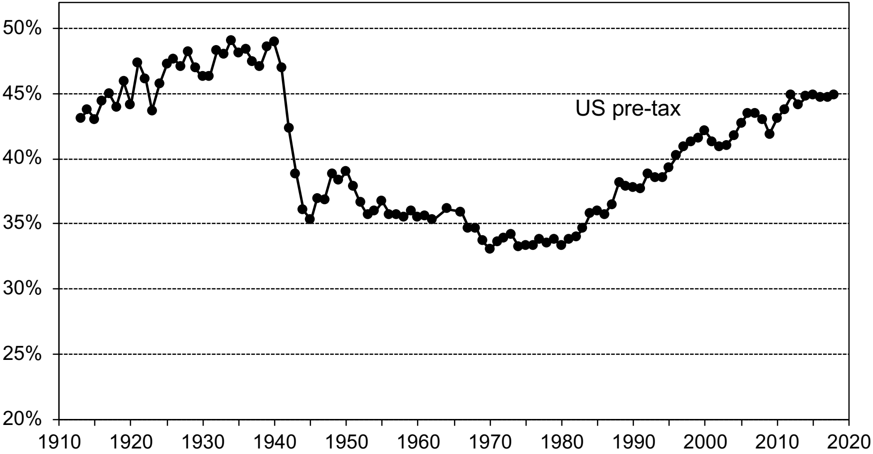
Main Market Failures

- **1) Externalities:** (example: greenhouse carbon emissions) $\Rightarrow$ require govt interventions (Pigouvian taxes/subsidies, public good provision)
- **2) Imperfect competition:** (example: monopoly) $\Rightarrow$ requires regulation (typically studied in Industrial Organization)
- **3) Imperfect or Asymmetric Information:** (example: adverse selection in health insurance may require mandatory insurance)
- **4) Individual failures:** People are not always rational. This is analyzed in behavioral economics, field in huge expansion (example: myopic people may not save enough for retirement)

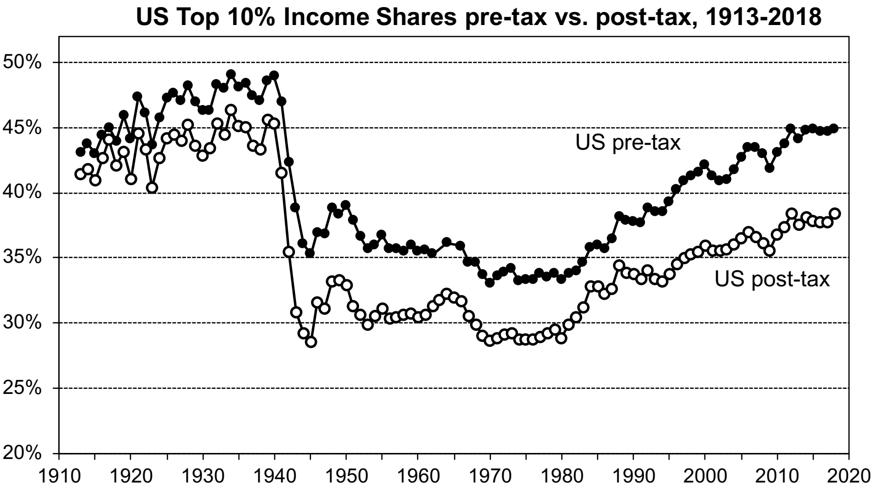
Inequality and Redistribution

- Even if market outcome is efficient, society might not be happy with the market outcome because market equilibrium might generate very high economic disparity across individuals
- Governments use taxes and transfers to redistribute from rich to poor and reduce inequality
- Redistribution through taxes and transfers might reduce incentives to work (**efficiency costs**)
- $\Rightarrow$ Redistribution creates an **equity-efficiency trade-off**
- Income inequality has soared in the United States in recent decades, and has moved to the forefront in the public debate (Piketty's 2014 book success, stats from Piketty-Saez-Zucman '18)

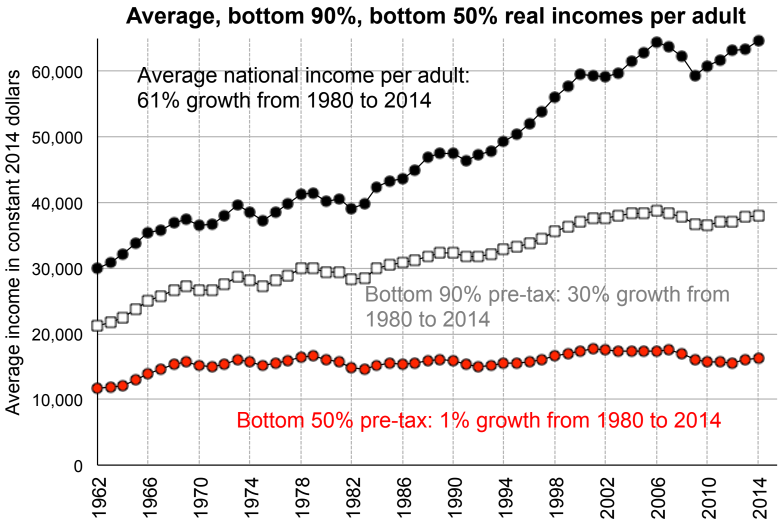
Top 10% Pre-tax Income Share in the US, 1913-2018

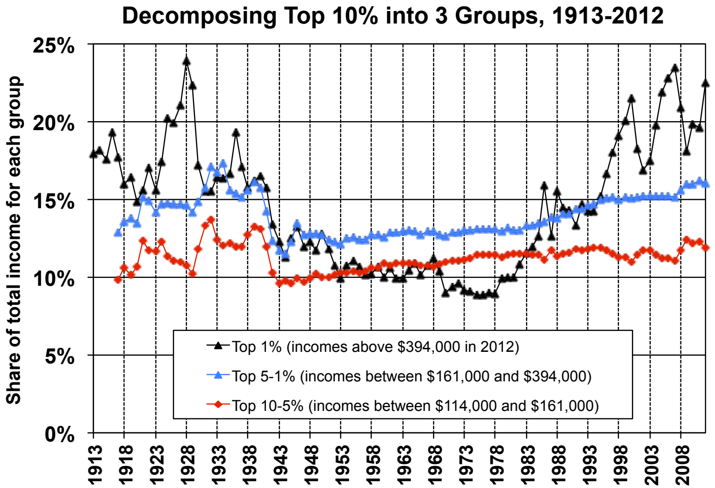


Top income shares of pretax national income among adults aged 20+ (income within couples equally split).
Source is World Inequality Database wid.world (from Piketty, Saez, Zucman 2018).



Top income shares of pretax and posttax national income among adults (income within married couples equally split). Source is Piketty, Saez, Zucman (2018) for US and Piketty et al. (2020) for France.


Average, bottom 90%, bottom 50% real incomes per adult
Average income in constant 2014 dollars
0
10,000
20,000
30,000
40,000
50,000
60,000
1962
1966
1970
1974
1978
1982
1986
1990
1994
1998
2002
2006
2010
2014
Average national income per adult:
61% growth from 1980 to 2014
Bottom 90% pre-tax: 30% growth from
1980 to 2014
Bottom 50% pre-tax: 1% growth from 1980 to 2014



Source: Piketty and Saez, 2003 updated to 2012. Series based on pre-tax cash market income including realized capital gains and excluding government transfers. 2012 data based on preliminary statistics.

How Might the Government Intervene?

- **1) Tax or Subsidize Private Sale or Purchase:** Tax goods that are overproduced (e.g. carbon tax) and subsidized goods underproduced (e.g., flu shots subsidies)
- **2) Restrict or Mandate Private Sale or Purchase:** Restrict the private sale or purchase of overproduced goods (e.g. fuel efficiency requirements), or mandate the private purchase of underproduced goods (e.g., auto insurance)
- **3) Public Provision:** The government can provide the good directly, in order to potentially attain the level of consumption that maximizes social welfare (example is National Defense)
- **4) Public Financing of Private Provision:** Government pays for the good but private sector supplies it (e.g., privately provided health insurance paid for by US government in Medicare-Medicaid)

What are the Effects of Alternative Interventions?

- **1) Direct Effects:** The effects of government interventions that would be predicted if individuals did not change their behavior in response to the interventions.
  Direct effects are relatively easy to compute
- **2) Indirect Effects:** The effects of government interventions that arise only because individuals change their behavior in response to the interventions (sometimes called **unintended effects**)
- Empirical public economics analysis tries to estimate indirect effects to inform the policy debate
- **Example:** increasing top income tax rates mechanically raises tax revenue but top earners might work less and earn less, reducing tax revenue relative to mechanical calculation

Why Do Governments Do What They Do?

- **Political economy**: The theory of how the political process produces decisions that affect individuals and the economy
- **Example:** Understanding how the level of taxes and spending is set through voting and voters' preferences
- **Public choice** is a sub-field of political economy from a Libertarian perspective that focuses on **government failures**
- Government failures = situations where the government does not act in the benefit of society (e.g., government captured by special interests or a self-perpetuating bureaucracy)

Normative vs. Positive Public Economics

- **Normative Public Economics:** Analysis of How Things Should be (e.g., should the government intervene in health insurance market? how high should taxes be?, etc.)
- **Positive Public Economics:** Analysis of How Things Really Are (e.g., Does govt provided health care crowd out private health care insurance? Do higher taxes reduce labor supply?)
- Positive Public Economics is a required 1st step before we can complete Normative Public Economics
- Positive analysis is primarily empirical and Normative analysis is primarily theoretical

Paternalism vs. Individual Failures

In many situations, individuals may not or do not seem to act in their best interests [e.g., many individuals are not able to save for retirement]

Two Polar Views on such situations:

(1) **Paternalism [Libertarian View]** Individual failures do not exist and government wants to impose its own preferences against individuals' will

(2) **Individual Failures [Behavioral Economics View]** Individual Failures exist: Self-control problems, Cognitive Limitations

Distinguishing the 2 views: Under Paternalism, individuals are opposed to government interventions. If individuals understand they have failures, they will support govt interventions.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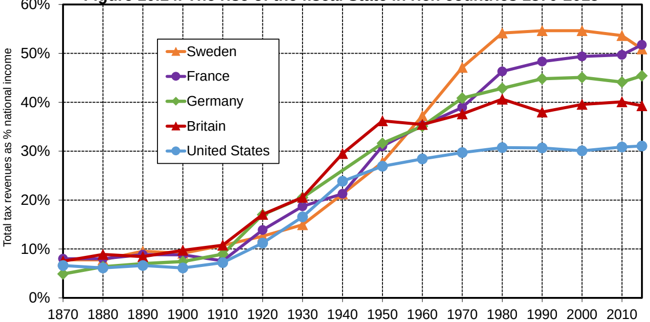
Key Facts on Taxes and Spending

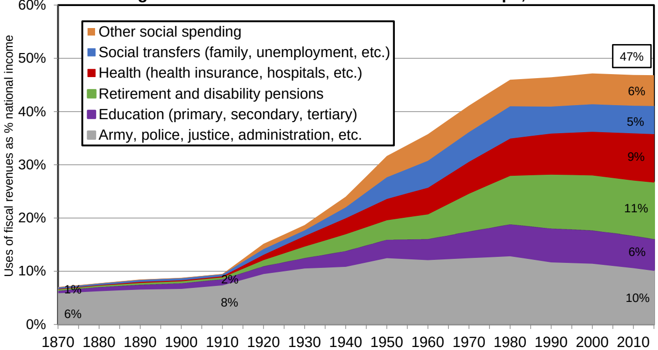
1. **Government Growth:** Size of government relative to National Income grows dramatically over the process of development from less than 10% in less developed economies to 30-50% in most advanced economies
2. **Government Size Stable** in richest countries after 1980
3. **Government Growth** is due to the expansion of the **welfare state:** public education, public retirement benefits, public health insurance, income support programs
4. **Govt spending** > **Taxes:** Most rich countries run deficits and have significant public debt (relative to GDP), particularly after Great Recession of 2008-10 and Covid 2020-21

Figure 10.14. The rise of the fiscal State in rich countries 1870-2015



Interpretation. Total fiscal revenues (all taxes and social contributions included) made less than 10% of national income in rich countries during the 19th century and until World War 1, before rising strongly from the 1910s-1920s until the 1970s-1980s and then stabilizing at different levels across countries: around 30% in the U.S., 40% in Britain and 45%-55% in Germany, France and Sweden.
Sources and series: see piketty.pse.ens.fr/ideology.

Figure 10.15. The rise of the social State in Europe, 1870-2015



Interpretation. In 2015, fiscal revenues represented 47% of national income on average in Western Europe et were used as follows: 10% of national income for regalian expenditure (army, police, justice, general administration, basic infrastructure: roads, etc.); 6% for education; 11% for pensions; 9% for health; 5% for social transfers (other than pensions); 6% for other social spending (housing, etc.). Before 1914, regalian expenditure absorbed almost all fiscal revenues. **Note.** The evolution depicted here is the average of Germany, France, Britain and Sweden (see figure 10.14). Sources and séries: see piketty.pse.ens.fr/ideology.

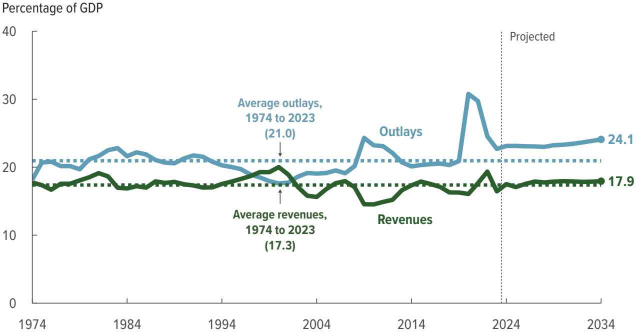
Different Levels of Governments

- US Federal govt raises about 20% of GDP in taxes
- State+Local govts raise about 10% of GDP in taxes
- Decentralized govt = a larger fraction of taxes/spending are decided at local level
- Decentralized govt can tailor policy to local views (example: California has more liberal policies than Texas)
- Redistribution through taxes and transfers harder to achieve at local level (rich can leave local jurisdiction if local taxes are too high) $\Rightarrow$ Local govts tend to do less redistribution
- $\Rightarrow$ Conservatives/libertarians tend to prefer decentralized states

Figure 1-3.

Total Federal Outlays and Revenues

Federal goverment only (state+local excluded)



Measured as a percentage of GDP, projected outlays remain about the same for the next several years as growth in outlays for interest payments is offset by decreases in discretionary spending. Over the 2024–2034 period, outlays exceed their 50-year average by more than revenues exceed their historical average.

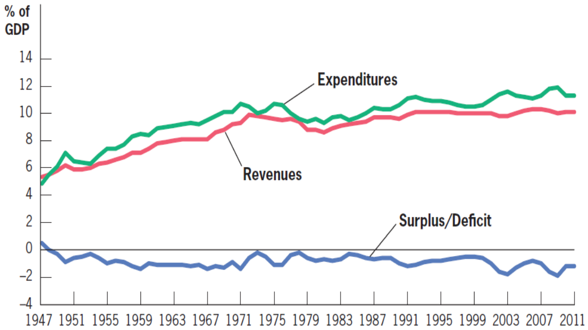
Data source: Congressional Budget Office. See www.cbo.gov/publication/59710#data.

When October 1 (the first day of the fiscal year) falls on a weekend, certain payments that would have ordinarily been made on that day are instead made at the end of September and thus are shifted into the previous fiscal year. All projections presented here have been adjusted to exclude the effects of those timing shifts. Historical amounts have been adjusted as far back as the available data will allow.

GDP = gross domestic product.

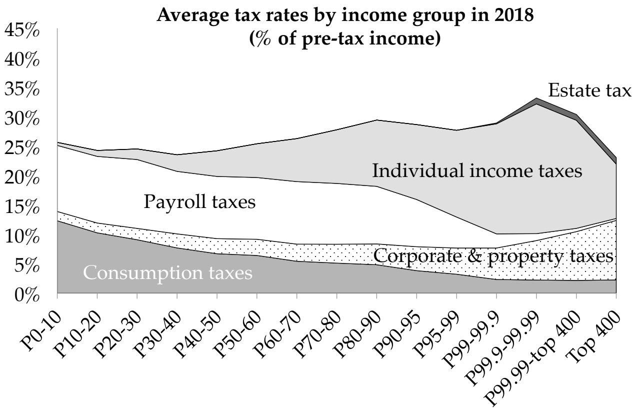
1.2

State and Local Government Receipts, Expenditures, and Surplus, 1947−2008



Distribution of Taxes

- US Federal govt raises about 20% of GDP in taxes, State+Local govt raises about 10% of GDP in taxes.
- Main Federal taxes: (1) Individual income tax (40%), (2) payroll taxes on earnings (40%), (3) corporate tax (15%)
- Main State taxes: (1) real estate property taxes (30%), (2) sales and excise taxes (30%), (3) individual and corporate state taxes (30%)
- Key questions: who bears the burden of those taxes (tax incidence), what impact do they have on the economy?



Source: Saez and Zucman (2019)

Regulatory Role of the Government

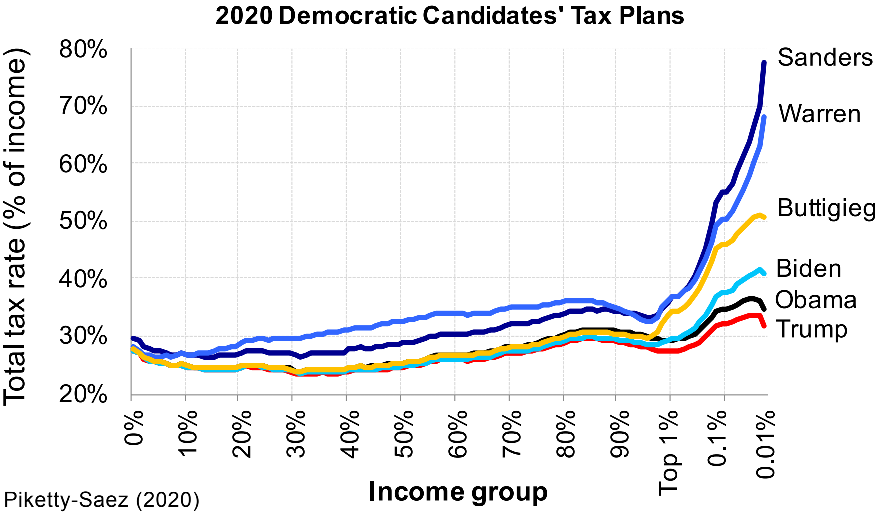
Another critical role the government plays in all nations is that of *regulating economic and social activities*.

Examples:

1. **Minimum wage** at the Federal level is $7.25 (States can adopt higher min wages) $\Rightarrow$ Potential impact on inequality
2. The **Food and Drug Administration (FDA)** regulates the labeling and safety of nearly all food products and approves drugs and medical devices to be sold to the public
3. The **Occupational Safety and Health Administration (OSHA)** is charged with regulating the workplace safety of American workers
4. The **Environmental Protection Agency (EPA)** is charged with minimizing dangerous pollutants in the air, water, and food supplies

Public Debates over Taxes, Health Care, and Climate Change

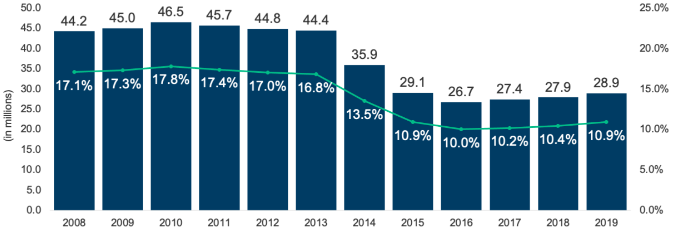
- Taxes, health care, and climate change are each the subject of debate, with both the "liberal" and "conservative" positions holding differing views in their approach to each problem.
    1. **Taxes:** U.S. decreased taxes on corporations and individuals in 2018 and 2025. Many want to tax the rich more. Many want to tax the middle class more.
    2. **Health Care:** Up to 2013, 17-18% of the non-elderly U.S. population not insured. With Obamacare down to 10%. Some propose Medicare for All. Some propose less government intervention.
    3. **Climate change:** Carbon emissions are generating global warming with potentially devastating future consequences (sea rise, extreme weather, agricultural output). What should government do?



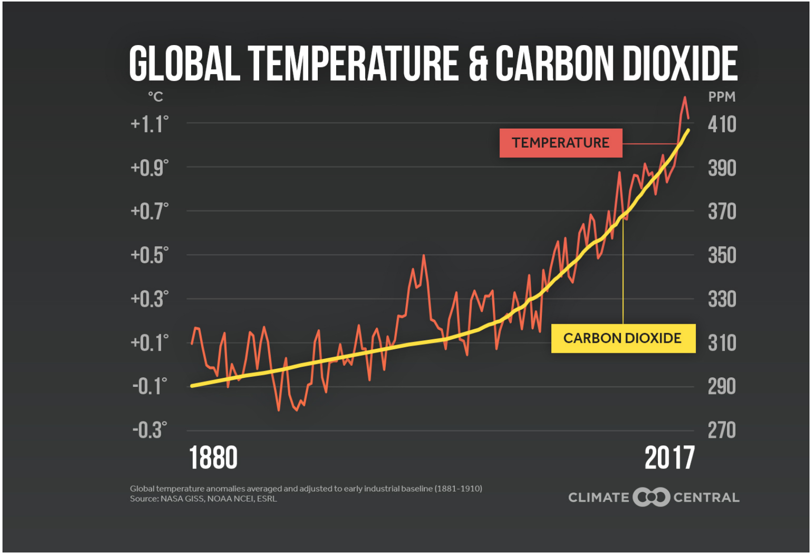
Piketty-Saez (2020)

Figure 1

Number of Uninsured and Uninsured Rate among the Nonelderly Population, 2008-2019



NOTE: Includes nonelderly individuals ages 0 to 64.
SOURCE: KFF analysis of 2008-2019 American Community Survey, 1-Year Estimates.

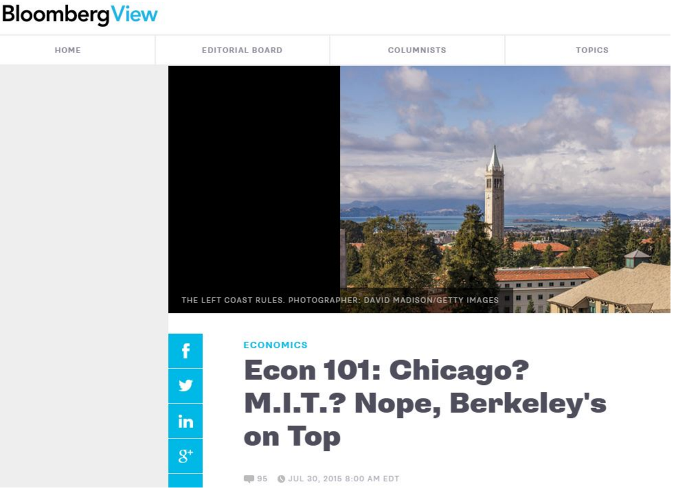

GLOBAL TEMPERATURE & CARBON DIOXIDE
°C
+1.1°
+0.9°
+0.7°
+0.5°
+0.3°
+0.1°
-0.1°
-0.3°
PPM
410
390
370
350
330
310
290
270
TEMPERATURE
CARBON DIOXIDE
1880
2017
Global temperature anomalies averaged and adjusted to early industrial baseline (1881-1910)
Source: NASA GISS, NOAA NCEI, ESRL
CLIMATE CENTRAL

Instruction at a Top Research University

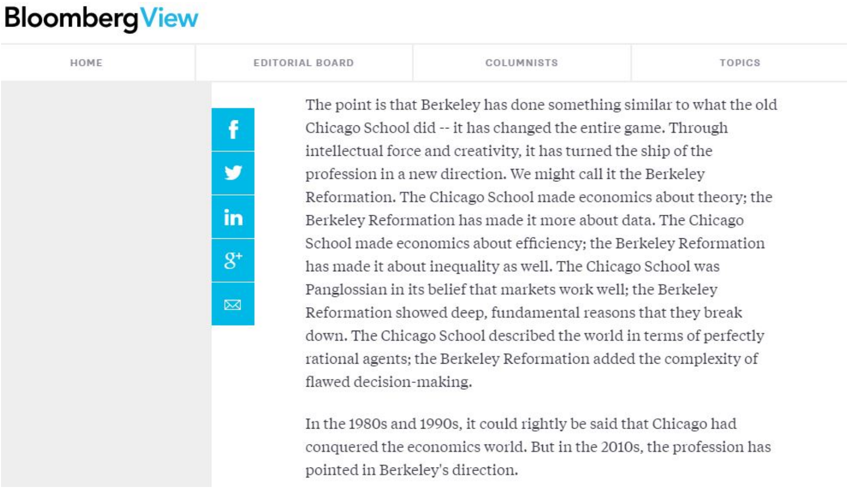
Three things you can get out of a course:

1. Information: Facts and tools
2. Perspective: What really matters and why
3. Inspiration: Belief in the power of ideas and hard work

Berkeley Economics: An Emphasis on Data



Berkeley Economics: An Emphasis on Data



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The point is that Berkeley has done something similar to what the old Chicago School did -- it has changed the entire game. Through intellectual force and creativity, it has turned the ship of the profession in a new direction. We might call it the Berkeley Reformation. The Chicago School made economics about theory; the Berkeley Reformation has made it more about data. The Chicago School made economics about efficiency; the Berkeley Reformation has made it about inequality as well. The Chicago School was Panglossian in its belief that markets work well; the Berkeley Reformation showed deep, fundamental reasons that they break down. The Chicago School described the world in terms of perfectly rational agents; the Berkeley Reformation added the complexity of flawed decision-making.

In the 1980s and 1990s, it could rightly be said that Chicago had conquered the economics world. But in the 2010s, the profession has pointed in Berkeley's direction.

We Will Follow That Tradition

- This course: Data-driven approach to what government intervention really does to economic efficiency and inequality and what really matters for policy decisions
- Along the way: Convey my confidence in you to use your ideas, reason, and hard work to improve lives

References

- Jonathan Gruber, Public Finance and Public Policy, Fourth Edition, 2012 Worth Publishers, Chapter 1
- National Center for Education Statistics "Highlights from TIMSS 2007: Mathematics and science achievement of US fourth-and eighth-grade students in an international context." Institute of Education Sciences, US Department of Education, 2009.**[web link]**
- Piketty, Thomas, *Capital in the 21st Century*, Cambridge: Harvard University Press, 2014, Chapter 13, **[web link]**
- Piketty, Thomas, *Capital and Ideology*, Cambridge: Harvard University Press, 2020, Chapter 10, **[web link]**
- Piketty, Thomas and Emmanuel Saez "Income Inequality in the United States, 1913-1998", Quarterly Journal of Economics, 118(1), 2003, 1-39, series updated to 2012 in September 2013 **[web link]**
- Piketty, Thomas, Emmanuel Saez, and Gabriel Zucman, "Distributional National Accounts: Methods and Estimates for the United States", Quarterly Journal of Economics, 133(2), 553-609, 2018 **[web link]**
- Saez, Emmanuel and Gabriel Zucman. The Triumph of Injustice: How the Rich Dodge Taxes and How to Make them Pay, New York: W.W. Norton, 2019. **[web link]**